

Transcriptomic Analysis of Organ-Specific Metastatic Adaptation in Breast Cancer

Background:

Metastasis is a context-dependent process influenced by both tumor-intrinsic properties and the target organ microenvironment. Understanding how cancer cells adapt transcriptionally to different metastatic niches remains a key challenge.

Methods:

Public microarray data (GSE62598) from organ-tropic metastatic breast cancer sublines (mouse 4T1 model) were analyzed. Probe-level data were mapped to gene symbols and collapsed to gene-level expression. PCA was performed on the top 1000 most variable genes. Differential expression analysis and pathway enrichment (KEGG) were conducted.

Key Findings:

- PCA showed a spread of liver-tropic samples along a dominant transcriptional axis, though with only $n=3$ per group this reflects a pattern, not a statistically established separation.
- Liver and lung metastatic sublines showed a narrower range of fold changes than liver vs bone, consistent with a smaller transcriptional distance between the former pair.
- No genes passed strict statistical thresholds ($FDR < 0.05$) due to small sample size.
- Exploratory analysis identified hundreds of differentially expressed genes under relaxed, uncorrected thresholds (373 in liver vs lung, 708 in liver vs bone).
- Pathway enrichment repeatedly surfaced metabolic processes — including lipid, amino acid, and central carbon metabolism — across comparisons, but these results did not reach significance after correction.

Interpretation:

These results do not establish that metastatic adaptation is driven by pathway-level rather than gene-level changes — the sample size is too small for that claim. What they do show is a recurring, non-significant enrichment signal in metabolic pathways across multiple comparisons, which is a reasonable basis for a follow-up hypothesis: that metabolic adaptation contributes to organ-specific colonization. Testing that hypothesis would require a larger, adequately powered dataset.

Conclusion:

This analysis demonstrates a full differential expression and enrichment pipeline applied to an organ-tropic metastasis model, and generates a specific, testable hypothesis — a role for metabolic adaptation in organ-specific colonization — for validation in future work with larger sample sizes.